V.L Memorial Public School

475 Vijay Nagar

CHAPTER-9-QUESTIONS-CLASS7TH-SCIENCE

Class 07 - Science (Curiosity)

Section A

1. Which animals are called ruminants? **[1]**

   a) Animals that bring back swallowed food and chew it again
   b) Animals that filter food
   c) Animals that swallow prey
   d) Animals that lack teeth

2. What is the process of inhaling and exhaling air called? **[1]**

   a) Gas exchange
   b) Energy release
   c) Breathing process
   d) Food digestion

3. What is the function of alveoli in the lungs? **[1]**

   a) Produce enzymes
   b) Filter waste
   c) Pump blood
   d) Exchange gases

4. What is the approximate length of the small intestine? **[1]**

   a) About 9 metres
   b) About 3 metres
   c) About 1.5 metres
   d) About 6 metres

5. The finger-like outgrowths of Amoeba help to ingest food. However, the finger-like outgrowths of the human intestine help to **[1]**

   a) digest the fatty food substances
   b) make the food soluble
   c) absorb the digested food
   d) absorb the undigested food.

6. Why is the small intestine longer than the large intestine? **[1]**

   a) To aid in respiration
   b) To store waste
   c) To increase surface area for absorption
   d) To produce digestive enzymes

7. Which of the following is a waste product of respiration? **[1]**

   a) Oxygen
   b) Carbon dioxide
   c) Water
   d) Glucose

8. Which of the following is NOT part of the digestive system? **[1]**

   a) Pancreas
   b) Kidney
   c) Stomach
   d) Oesophagus

9. Which of the following is not a component of the circulatory system? **[1]**

a) Blood
b) Blood vessels
c) Lungs
d) Heart

10. What is the process called in which complex food components are broken down into simpler forms? **[1]**

a) Digestion of food
b) Absorption of nutrients
c) Respiration of nutrients
d) Egestion of waste

11. What happens to the ribs during exhalation? **[1]**

a) They remain raised
b) They move upward
c) They move outward
d) They move downward

12. During inhalation, the diaphragm: **[1]**

a) Closes the windpipe
b) Reduces space in the lungs
c) Contracts and moves down
d) Relaxes and moves up

13. Which process moves digested nutrients from the small intestine into the blood? **[1]**

a) Waste egestion
b) Mechanical digestion
c) Food movement
d) Nutrient absorption

14. Which process provides energy to animals? **[1]**

a) Excretion
b) Reproduction
c) Circulation
d) Respiration

15. Which organ secretes pancreatic juice during digestion? **[1]**

a) Large intestine
b) Stomach
c) Pancreas
d) Liver

16. Which digestive secretion is mildly basic in nature? **[1]**

a) Bile
b) Stomach secretion
c) Saliva
d) Mouth secretion

17. What is the main function of the large intestine? **[1]**

a) Digest proteins
b) Absorb water
c) Absorb nutrients
d) Produce bile

18. Which organ filters waste from the blood to form urine? **[1]**

a) Lungs
b) Kidney
c) Liver
d) Heart

19. Which substances are produced when glucose undergoes respiration? **[1]**

a) Oxygen and glucose
b) Glucose and oxygen
c) Carbon dioxide and water
d) Water and glucose

20. What is the role of saliva in digestion? **[1]**

a) It neutralises stomach acid.
b) It digests proteins.

c) It breaks down fats.

d) It moistens food and digests starch.

21. **Assertion(A):** Inhalation is caused by the expansion of the chest cavity. **[1]**
Reason (R): The diaphragm contracts during inhalation.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

22. **Assertion (A):** In humans, respiration occurs in the lungs. **[1]**
Reason (R): Alveoli in the lungs provide a large surface area for gas exchange.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

23. **Assertion (A):** The nose filters and moistens the air we breathe. **[1]**
Reason (R): Tiny hairs and mucus help trap dust particles.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

24. **Assertion (A):** Bile helps in the digestion of fats. **[1]**
Reason (R): Bile emulsifies fats, breaking them into smaller droplets.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

25. **Assertion (A):** The small intestine is the longest part of the digestive system. **[1]**
Reason (R): It is the main site for digestion of proteins.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

26. **Assertion (A):** Anaerobic respiration in muscles produces alcohol. **[1]**
Reason (R): Alcohol fermentation occurs in yeast, not in humans.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

27. **Assertion (A):** The pancreas is part of the respiratory system. **[1]**
Reason (R): The pancreas produces pancreatic juice that is basic in nature.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

28. What is the role of bile? **[1]**

29. What is the function of the rectum in the digestive system? **[1]**

30. What is the function of the rumen in cows? **[1]**

31. Which organ secretes the juice that helps neutralise the acid? **[1]**

32. What is rumination? **[1]**

33. What is Tummo breathing? **[1]**

34. What is digestion? Where does it begin in humans? **[1]**

Section B

35. Match the following: **[2]**

Column (A)	Column (B)
(a) Mouth	(i) Produces bile
(b) Small Intestine	(ii) Digests proteins using acids
(c) Stomach	(iii) Main site of nutrient absorption
(d) Liver	(iv) Begins starch digestion

36. Match the following: **[2]**

Column A	Column B
(a) Aerobic respiration	(i) Occurs without oxygen
(b) Anaerobic respiration	(ii) Produces less energy
(c) Glucose	(iii) A sugar used as fuel in respiration
(d) Oxygen	(iv) Required for aerobic respiration

37. Explain how hair and mucus in the nostrils help during breathing. **[2]**

38. How do aquatic animals like fish breathe? **[2]**

39. Why do frogs have moist skin, and how does it help them? **[2]**

40. How does respiration occur in insects? **[2]**

41. Sahil placed some pices of chapati in test tube A. Neha placed chewed chapati in test tube B, and Santushti took boiled and mashed potato in test tube C. All of them added a few drops of iodine solution to their test tubes A, B, and C, respectively. What would be their observations? Give reasons. **[2]**

42. Describe the process of rumination in animals. **[2]**

43. Describe the benefits of Pranayama and Tummo breathing practices. **[2]**

Section C

44. How does the human digestive system differ from that of an insect (e.g., grasshopper)? **[3]**

45. Why do frogs have both lungs and moist skin for respiration? Explain. **[3]**

46. Explain how oxygen and carbon dioxide are exchanged through the alveoli. **[3]**

47. Why are acid and mucus important during digestion in the stomach? **[3]**

48. Explain why fish die when taken out of water, even if oxygen is present in air. **[3]**

49. What does the iodine test show about the action of saliva on starch in chewed rice? **[3]**

50. Compare the circulatory systems of fish and birds. How do these differences help **[3]**

them survive?

Section D

51. **Read the following text carefully and answer the questions that follow:** **[4]**

After most of the nutrients are digested and absorbed in the small intestine it moves into the large intestine. The large intestine is about 1.5 metres in length. It is shorter than the small intestine. The reason is that it is wider than the small intestine. The large intestine absorbs water and some salts from the undigested food, thus making the waste semi-solid. This semi-solid waste is called stool. The stool is then stored in the lower part of the large intestine, called the rectum, until the body is ready to get rid of it. Eating fibre-rich foods like fruits, vegetables, and whole grains helps the large intestine function properly by making the stool easier to pass. Finally, it is expelled through the anus-a process known as egestion.

Grass-eating animals, such as cows and buffaloes, partially chew the grass and swallow it into their stomachs. In the stomach, partial digestion of the food takes place. The partially digested food is brought back to the mouth for gradual chewing. This process is called rumination, and these animals are called ruminants. A cow spends about 8 hours a day just chewing the food! The thoroughly chewed food again passes down the alimentary canal for further digestion. Birds do not have teeth, but they have a chamber called a gizzard. Food is broken down by the contraction and relaxation of the walls of the gizzard, often with the help of grit (small stones) that the birds swallow.

Questions:

a. What is the function of the rectum in the human digestive system? (1)

b. What is the name of the process by which birds break down food in the absence of teeth? (1)

c. What role does the large intestine play in the digestive process? (2)

OR

Who are ruminants and what is meant by rumination? (2)

52. **Read the following text carefully and answer the questions that follow:** **[4]**

When you chew your food, your saliva not only helps in digesting the starch but also moistens it, making it soft and easy to swallow. Your tongue helps in mixing chewed food with saliva and pushing this softened food into a long, flexible tube called the food pipe or oesophagus. The walls of the food pipe gently contract and relax in a wave-like motion to push the food down into the stomach. This movement takes place throughout the alimentary canal and pushes the food forward. In the stomach, the walls contract and relax to churn the food. The churned food is then mixed with a secretion from the inner lining of the stomach. The secretion from stomach contains digestive juice, acid, and mucus. The digestive juice of the stomach breaks down proteins present in the food into simpler components. The acid not only helps break down proteins but also kills many harmful bacteria. The mucus protects the stomach lining from the acid, preventing damage. In the stomach, the food is partially

digested and transformed into a semi-liquid mass, preparing it for the next stage of digestion.

Questions:

a. What is the function of saliva during chewing of food? (1)

b. Name the wave-like movement that pushes food through the food pipe. (1)

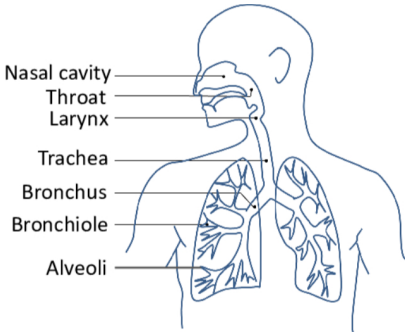
c. What role does the stomach play in digestion? Mention the components of its secretion. (2)

OR

Explain how food moves from the mouth to the stomach. (2)

53. **Read the following text carefully and answer the questions that follow:** **[4]**

We know that we breathe in (inhale) and breathe out (exhale) air continuously to obtain oxygen and release carbon dioxide. The process of inhaling and exhaling air is called breathing. It is difficult to live without food for a week; without water for a day or two, but without breathing, we usually cannot survive more than a few minutes. This system is called respiratory system.



The respiratory system consists of various parts. In this system, the exchange of gases follows a specific pathway. The pathway through which the air is inhaled and exhaled involves various parts of the respiratory system assisting in the process of breathing and respiration. The respiratory system begins with a pair of nasal openings called nostrils through which we inhale and exhale air. The inhaled air passes into a pair of small passages called the nasal passages. We have noticed tiny hair inside our nostrils, these hair along with mucus, help trap dust and dirt from the air we breathe in. This is why we should breathe through the nose and not through the mouth. From the nasal passages, the air reaches our lungs through the windpipe. The windpipe forms two branches, which enter the two lungs. In the lungs, these branches further divide into smaller and finer branches that end in small balloon-like sacs called alveoli.

Questions:

a. What is the function of the tiny hair and mucus present in the nasal passages? (1)

b. What is the name of the balloon-like sacs present in the lungs where exchange of gases occurs? (1)

c. Why is it advised to breathe through the nose and not through the mouth? (2)

OR

Describe the path of air from the nostrils to the lungs in the human respiratory system. (2)

Section E

54. How do different animals use different feeding methods to obtain food? **[5]**

55. Compare and contrast the respiratory systems of frogs, fish, and humans. How are they adapted to their habitats? **[5]**

56. Why do birds require a high-energy diet? Explain how their digestive and circulatory systems support flight. **[5]**

57. Analyze how ruminants and birds show different adaptations for breaking down food. **[5]**

58. Analyze how different breathing mechanisms help animals survive in their habitats. **[5]**

59. Describe how different animals (amoeba, hydra, earthworm, and human) excrete waste. Compare their methods. **[5]**

Section F

60. **State whether the given statement is True or False:** **[8]**

    (a) Excretion is the process of removing undigested food from the body. **[1]**

    (b) Digestion is the process of breaking down food into simpler substances. **[1]**

    (c) We should always breathe through the mouth and not the nose. The ribcage protects our lungs. **[1]**

    (d) The liver produces bile that helps in the digestion of proteins. **[1]**

    (e) Respiration is the process by which food is broken down to release energy. **[1]**

    (f) The heart pumps blood throughout the body, supplying oxygen and nutrients. **[1]**

    (g) The stomach is the site where all food is completely digested. **[1]**

    (h) The ruminants bring back swallowed grass into their mouth and chew it for some time. **[1]**

61. **Fill in the blanks:** **[8]**

    (a) The final stage of digestion, where undigested material is removed, is called _____. **[1]**

    (b) The process by which animals take in oxygen and release carbon dioxide is called _____. **[1]**

    (c) The waste product exhaled during respiration is _____. **[1]**

    (d) Respiration in earthworms takes place through their _____. **[1]**

    (e) The body part where undigested food is stored temporarily before being removed is the _____. **[1]**

    (f) The process in which energy is released from food using oxygen is called _____ respiration. **[1]**

    (g) The fluid part of blood that carries nutrients and hormones is called _____. **[1]**

    (h) The respiratory organs in fish are called _____. **[1]**